

INDIAN INSTITUTE OF TECHNOLOGY, GUWAHATI

This report presents the ongoing work carried out as part of the

M.Tech Project

on

Bearing Based Control Of Quadrotors in GPS Denied Environment

The project is currently in progress.

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1 Introduction

This M.Tech project focuses on the development and analysis of a **target–leader tracking control system** for achieving accurate position tracking and bearing alignment between a leader and a moving target. The objective is to enable the leader to maintain a prescribed distance from the target while following its desired direction of motion.

The project considers the mathematical modeling of the target and leader using a single-integrator dynamic model. A control law is developed using **position and bearing feedback** to reduce the tracking error and align the actual bearing with the desired bearing. The resulting closed-loop error dynamics are derived and analyzed using **Lyapunov stability theory**.

The project further investigates the effect of a time-varying desired bearing on system stability. An **Input-to-State Stability (ISS)** framework is used to characterize the influence of the changing reference trajectory on the tracking errors. The ongoing work includes mathematical analysis, controller development, stability verification, and simulation-based evaluation of the proposed control strategy.

Overall, the project aims to develop a reliable control approach that enables the leader to follow a moving target while maintaining the desired spatial relationship and demonstrating stable tracking performance.

2 Objectives

The main objectives of this M.Tech project are:

1. **To develop a mathematical model** of the target–leader system using a single-integrator dynamic model.
2. **To formulate the position tracking error** and establish a desired reference position for the leader at a prescribed distance behind the moving target.
3. **To design a cooperative control law** using position and bearing feedback to achieve accurate tracking of the desired reference position.
4. **To align the actual bearing vector** of the leader with the desired bearing direction associated with the target’s motion.
5. **To derive the complete closed-loop error dynamics** for both position tracking and bearing alignment.
6. **To analyze the stability of the proposed control system** using Lyapunov stability theory.
7. **To investigate the effect of a time-varying desired bearing** on the tracking performance and system stability.
8. **To establish an Input-to-State Stability (ISS) framework** for analyzing the influence of the changing reference trajectory on the tracking errors.

9. **To validate the proposed control strategy through simulation** and evaluate its tracking, stability, and convergence performance.
10. **To develop a reliable and robust tracking framework** that enables the leader to maintain the desired spatial relationship with a moving target.

3 Problem Formulation

Consider a target–leader system moving in $\mathbb{R}^d$, where $d = 2$ or $d = 3$. The positions of the target and leader are denoted by $p_T(t), p_L(t) \in \mathbb{R}^d$, respectively. The dynamics of the target and leader are given by the single-integrator model

$$\dot{p}_T = v_T, \quad \dot{p}_L = u_L, \quad (1)$$

where $v_T \in \mathbb{R}^d$ is the target velocity and $u_L \in \mathbb{R}^d$ is the control input of the leader. The relative position vector between the target and the leader is defined as

$$r = p_T - p_L. \quad (2)$$

The corresponding relative distance is given by

$$d = \|r\|. \quad (3)$$

The current bearing vector from the leader to the target is defined as

$$g = \frac{r}{\|r\|} = \frac{p_T - p_L}{\|p_T - p_L\|}. \quad (4)$$

Thus, g represents the direction from the leader toward the target.

Let $g^*(t)$ denote the desired bearing vector. For a moving target, the desired bearing is selected along the direction of the target velocity, given by

$$g^*(t) = \frac{v_T(t)}{\|v_T(t)\|}, \quad \|v_T(t)\| \neq 0. \quad (5)$$

A prescribed distance $d_{TL} > 0$ is selected between the target and the leader. Accordingly, the desired reference position of the leader is defined as

$$p_{L,\text{ref}}(t) = p_T(t) - d_{TL}g^*(t). \quad (6)$$

Hence, the desired reference position is located d_{TL} units behind the target along the desired direction of motion.

The leader position tracking error is defined as

$$z = p_{L,\text{ref}} - p_L. \quad (7)$$

Substituting the reference position gives

$$z = p_T - p_L - d_{TL}g^*. \quad (8)$$

Using the relative position vector $r = p_T - p_L$, the tracking error can be written as

$$z = r - d_{TL}g^*. \quad (9)$$

The objective is to design a control law for the leader such that the position tracking error z converges to zero while simultaneously ensuring that the actual bearing vector g asymptotically aligns with the desired bearing vector g^* . In other words,

$$\lim_{t \rightarrow \infty} z(t) = 0 \quad (10)$$

and

$$\lim_{t \rightarrow \infty} (g(t) - g^*(t)) = 0. \quad (11)$$

Therefore, the leader is required to track a point located at a prescribed distance d_{TL} behind the target while maintaining the desired bearing relationship with respect to the target.

4 Closed-Loop Dynamics

4.1 Relative Position and Desired Reference Position

Let the relative position vector between the target and the leader be

$$r = p_T - p_L \quad (12)$$

where p_T denotes the target position and p_L denotes the leader position.

The corresponding relative distance is

$$d = \|r\| \quad (13)$$

The current bearing vector is defined as

$$g = \frac{r}{d} \quad (14)$$

The desired reference position for the leader is defined as

$$p_{L,\text{ref}}(t) = p_T(t) - d_{TL}g^*(t) \quad (15)$$

where the desired bearing vector is given by

$$g^*(t) = \frac{v_T(t)}{\|v_T(t)\|} \quad (16)$$

Here, $v_T(t)$ represents the target velocity and $g^*(t)$ represents the desired bearing direction.

The parameter d_{TL} denotes the prescribed distance between the target and the leader reference position. Therefore, the reference position of the leader is located d_{TL} units

behind the target along the desired direction of motion. The parameter d_{TL} is user input and can be adjusted based on the desired formation or spacing between the leader and the target. Thus, the leader is required to track the reference point

$$p_{L,\text{ref}}(t) = p_T(t) - d_{TL}g^*(t), \quad (17)$$

which is located at a prescribed distance d_{TL} behind the target and aligned with the desired direction of motion.

Let the leader position error be defined as

$$z = p_{L,\text{ref}} - p_L \quad (4)$$

$$= (p_T - d_{TL}g^*) - p_L \quad (18)$$

$$= (p_T - p_L) - d_{TL}g^* \quad (19)$$

$$z = r - d_{TL}g^* \quad (5)$$

$$z = p_{L,\text{ref}} - p_L \quad (20)$$

4.1.1 Position Error Dynamics

Assumption:

$$p_L \in C^1, \quad p_T \in C^1,$$

$$p_{L,\text{ref}} \in C^1,$$

$$\dot{p}_T = v_T \quad \text{exists,}$$

$$g^*(t) = \frac{v_T}{\|v_T\|}$$

Now assume $p_T \in C^2$

so, $\dot{g}^*(t)$ exists.

Differentiating this with time,

$$\dot{z} = \dot{p}_{L,\text{ref}} - \dot{p}_L \quad (6)$$

Now,

$$p_{L,\text{ref}} = p_T - d_{TL}g^* \quad (21)$$

Differentiating this with time,

$$\dot{p}_{L,\text{ref}} = \dot{p}_T - d_{TL}\dot{g}^* \quad (7)$$

Now, time derivative of target position,

$$\dot{p}_T = v_T \quad (8)$$

So,

$$\dot{p}_{L,\text{ref}} = v_T - d_{TL}\dot{g}^* \quad (9)$$

The proposed control law for the leader is

$$\dot{p}_L = v_L \quad (22)$$

$$\dot{p}_L = v_T + k_p z - k_b P(g)g^* \quad (23)$$

where, k_p and k_b are position and bearing feedback gains,

$$k_p, k_b > 0$$

and

$$P(g) = I - gg^T \quad (24)$$

Now,

$$\dot{z} = \dot{p}_{L,\text{ref}} - \dot{p}_L \quad (25)$$

$$= \dot{p}_T - d_{TL}\dot{g}^* - \dot{p}_L \quad (26)$$

$$= (v_T - d_{TL}\dot{g}^*) - (v_T + k_p z - k_b P(g)g^*) \quad (27)$$

$$\boxed{\dot{z} = -k_p z + k_b P(g)g^* - d_{TL}\dot{g}^*} \quad (28)$$

$\Rightarrow$ this is exact position error dynamics

4.1.2 Bearing Error Dynamics

Current Bearing

$$g = \frac{r}{d}$$

where

$$r = p_T - p_L$$

and

$$d = \|r\| > 0$$

Take time derivative of g ,

$$\dot{g} = \frac{\dot{r}d - r\dot{d}}{d^2} \quad (29)$$

$$= \frac{\dot{r}}{d} - \frac{r}{d^2}\dot{d} \quad (30)$$

$$d = \|r\| = \sqrt{r^T r}$$

$$\dot{d} = \frac{1}{2}(r^T r)^{-\frac{1}{2}} \frac{d}{dt}(r^T r) \quad (31)$$

$$= \frac{1}{2d} (\dot{r}^T r + r^T \dot{r}) \quad (32)$$

Since

$$\dot{r}^T r = r^T \dot{r}$$

(as both of them is scalar) because r is column vector,

$$\dot{d} = \frac{r^T \dot{r}}{d} \quad (33)$$

Now, from the definition of g ,

$$g = \frac{r}{d}$$

$$r = dg$$

$$\dot{d} = \frac{(dg)^T r}{d} \quad (34)$$

$$\dot{d} = \dot{g}^T r$$

$$\dot{g} = \frac{\dot{r}}{d} - \frac{r}{d^2}\dot{d} \quad (35)$$

$$= \frac{\dot{r}}{d} - \frac{r}{d^2} (g^T \dot{r}) \quad (36)$$

$$= \frac{\dot{r}}{d} - \frac{dg}{d^2} (g^T \dot{r}) \quad (37)$$

$$= \frac{1}{d} (I - gg^T) \dot{r} \quad (38)$$

$$\dot{g} = \frac{1}{d}P(g)\dot{r} \quad (39)$$

$$P(g) = I - gg^T$$

now, from the definition of r ,

$$r = p_T - p_L \quad (40)$$

Differentiating with time,

$$\dot{r} = \dot{p}_T - \dot{p}_L \quad (41)$$

$$\dot{p}_T = v_T \quad (42)$$

Therefore,

$$\dot{r} = v_T - \dot{p}_L \quad (43)$$

The proposed control law,

$$\dot{p}_L = v_T + k_p z_{LT} - k_b P(g)g^* \quad (44)$$

Therefore,

$$\dot{r} = v_T - (v_T + k_p z_{LT} - k_b P(g)g^*) \quad (45)$$

$$\boxed{\dot{r} = -k_p z_{LT} + k_b P(g)g^*} \quad (46)$$

now put the value of $\dot{r}$ in the below equation that is obtained from equation 28

$$\dot{g} = \frac{1}{d}P(g)\dot{r} \quad (47)$$

$$= \frac{1}{d}P(g) [-k_p z_{LT} + k_b P(g)g^*] \quad (48)$$

$$\dot{g} = -\frac{k_p}{d}P(g)z_{LT} + \frac{k_b}{d}P(g)^2 g^* \quad (49)$$

now, we will prove that the projection matrix is idempotent.

$$\boxed{P(g)^2 = P(g)}$$

Proof

$$P(g)^2 = (I - gg^T)(I - gg^T)$$

$$= I - gg^T - gg^T + gg^T gg^T$$

$$\underbrace{g^T g}_{=1}$$

So,

$$gg^T gg^T = gg^T$$

$$P(g)^2 = I - 2gg^T + gg^T$$

$$= I - gg^T$$

$$= P(g)$$

So,

$$P(g)^2 = P(g)$$

So, the projection matrix is idempotent.

$$\text{Now, put } P(g)^2 = P(g)$$

in the equation 49, we get

$$\dot{g} = -\frac{k_p}{d}P(g)z_{TL} + \frac{k_b}{d}P(g)g^*$$

$$\dot{g} = -\frac{k_p}{d}P(g)z_{TL} + \frac{k_b}{d}P(g)g^*$$

So, the complete closed loop error system

$$\begin{aligned}\dot{z} &= -k_p z + k_b P(g)g^* - d_{TL}\dot{g}^* \\ \dot{g} &= -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^*\end{aligned}$$

The first equation governs the evaluation of the proposed position tracking error, while the second describes the evaluation of the bearing error.

5 Lyapunov Stability Analysis

The proposed control law

$$\dot{p}_L = u_L = v_T + k_p z - k_b P(g)g^* \quad (50)$$

is designed to accomplish two control objectives simultaneously:

- To drive the leader toward its desired reference position.
- To align the current bearing vector with the desired bearing direction.

Consequently, the Lyapunov function should quantify both the position tracking error and the bearing error within a single scalar energy function. To achieve this, we construct a composite Lyapunov function by combining an energy associated with the position error and another energy associated with the angular misalignment between the current and desired bearing vector.

So, let

$$V = V_p + CV_b \quad (51)$$

$$V = \frac{1}{2}z^T z + C(1 - g^T g^*) \quad (52)$$

where V_p is the position error energy, V_b is the bearing error energy and $C > 0$ is a positive constant that weights the relative importance of the

5.1 postive definiteness of Lyapunov function

5.1.1 positive definiteness of position error energy

$$V_p = \frac{1}{2}z^T z \quad (53)$$

$$z = p_{L,\text{ref}} - p_L$$

The quantity V_p represents the squared distance between the leader and its desired reference position. It satisfies the following fundamental properties,

$$V_p \geq 0 \quad \forall z \neq 0$$

and

$$V_p = 0 \iff z = 0$$

So, V_p is positive definite with the position error.

5.1.2 positive definiteness of bearing error energy

$$V_b = 1 - g^T g^* \tag{54}$$

$$\begin{aligned} \|g - g^*\|^2 &= (g - g^*)^T (g - g^*) \\ &= g^T g + g^{*T} g^* - 2g^T g^* \end{aligned}$$

$$g^T g = g^{*T} g^* = 1$$

$$\therefore \|g - g^*\|^2 = 2 - 2g^T g^*$$

$$= 2(1 - g^T g^*)$$

Since

$$V_b = 1 - g^T g^*$$

and

$$\|g - g^*\|^2 = 2(1 - g^T g^*),$$

we obtain

$$V_b = \frac{1}{2} \|g - g^*\|^2$$

Therefore,

$$g \neq g^* \implies \|g - g^*\|^2 > 0 \implies V_b > 0.$$

Similarly,

$$g = g^* \implies \|g - g^*\|^2 = 0 \implies V_b = 0.$$

Hence,

$$V_b > 0 \quad \forall g \neq g^*$$

and

$$V_b = 0 \iff g = g^*$$

Thus, V_b is positive definite with respect to the bearing error for $C > 0$.

$$V = V_p + CV_b$$

and we can see

$$V = 0$$

if

$$V_p = 0 \quad \text{and} \quad V_b = 0$$

which implies

$$z = 0$$

and

$$g = g^*.$$

and

$$V > 0$$

for

$$z \neq 0 \quad \text{and} \quad g \neq g^*.$$

So,

The proposed Lyapunov function is

$$V(z, g, g^*) = \frac{1}{2}\|z\|^2 + C(1 - g^T g^*), \quad C > 0. \quad (55)$$

Since g and g^* are unit vectors,

$$1 - g^T g^* = \frac{1}{2}\|g - g^*\|^2. \quad (56)$$

Therefore,

$$\boxed{V = \frac{1}{2}\|z\|^2 + \frac{C}{2}\|g - g^*\|^2} \quad (57)$$

and hence

$$V > 0 \quad (58)$$

for

$$(z, g - g^*) \neq (0, 0), \quad (59)$$

while

$$V = 0 \iff z = 0, \quad g = g^*. \quad (60)$$

Thus, V is positive definite with respect to the tracking-error vector

$$e = \begin{bmatrix} z \\ g - g^* \end{bmatrix}. \quad (61)$$

6 Derivative of Lyapunov Function

$$V = V_p + CV_b$$

$$= \frac{1}{2}z^T z + C(1 - g^T g^*)$$

Differentiating the position energy term

$$V_p = \frac{1}{2}z^T z$$

$$\dot{V}_p = \frac{1}{2}(\dot{z}^T z + z^T \dot{z})$$

Since $\dot{z}^T z$ and $z^T \dot{z}$ are scalars,

$$\dot{z}^T z = z^T \dot{z}$$

Therefore,

$$\dot{V}_p = z^T \dot{z}$$

Previously, we get position error dynamics

$$\dot{z} = -k_p z + k_b P(g)g^* - d_{TL}\dot{g}^*$$

So,

$$\dot{V}_p = z^T (-k_p z + k_b P(g)g^* - d_{TL}\dot{g}^*)$$

$$\dot{V}_p = -k_p \|z\|^2 + k_b z^T P(g)g^* - d_{TL} z^T \dot{g}^*$$

Differentiate the bearing energy

Consider

$$CV_b = C(1 - g^T g^*) \quad (62)$$

Therefore,

$$C\dot{V}_b = -C(\dot{g}^T g^* + g^T \dot{g}^*) \quad (63)$$

Previously, we get bearing error dynamics

$$\dot{g} = -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^* \quad (64)$$

Putting this in,

$$C\dot{V}_b = -C \left[\left(-\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^* \right)^T g^* + g^T \dot{g}^* \right] \quad (65)$$

Since

$$P(g)^T = P(g) \quad (66)$$

we get

$$C\dot{V}_b = -C \left[-\frac{k_p}{d}z^T P(g)g^* + \frac{k_b}{d}g^{*T} P(g)^T g^* + g^T \dot{g}^* \right] \quad (67)$$

Since

$$P(g) = I - gg^T \quad (68)$$

we have

$$C\dot{V}_b = -C \left[-\frac{k_p}{d}z^T P(g)g^* + \frac{k_b}{d}g^{*T} (I - gg^T) g^* + g^T \dot{g}^* \right] \quad (69)$$

$$= -C \left[-\frac{k_p}{d}z^T P(g)g^* + \frac{k_b}{d}(g^{*T}g^* - g^{*T}gg^Tg^*) + g^T \dot{g}^* \right] \quad (70)$$

Since

$$g^{*T}g^* = 1 \quad (71)$$

and Since both vectors are unit vectors,

$$\|g\| = \|g^*\| = 1$$

So, their inner product satisfies

$$g^T g^* = \cos \alpha$$

where

$$\alpha \in [0, \pi]$$

α is the angle between the two bearing vectors. so, we have

$$g^T g^* = \cos \alpha, \quad (72)$$

we obtain

$$C\dot{V}_b = -C \left[-\frac{k_p}{d} z^T P(g) g^* + \frac{k_b}{d} \left\{ 1 - (g^T g^*)^2 \right\} + g^T \dot{g}^* \right] \quad (73)$$

$$C\dot{V}_b = -C \left[-\frac{k_p}{d} z^T P(g) g^* + \frac{k_b}{d} (1 - \cos^2 \alpha) + g^T \dot{g}^* \right] \quad (74)$$

$$= -C \left[-\frac{k_p}{d} z^T P(g) g^* + \frac{k_b}{d} \sin^2 \alpha + g^T \dot{g}^* \right] \quad (75)$$

$$C\dot{V}_b = \frac{Ck_p}{d} z^T P(g) g^* - \frac{Ck_b}{d} \sin^2 \alpha - Cg^T \dot{g}^* \quad (76)$$

$$\begin{aligned} \dot{V} = & -k_p \|z\|^2 + \left(k_b + \frac{Ck_p}{d} \right) z^T P(g) g^* \\ & - \frac{Ck_b}{d} \sin^2 \alpha - d_{TL} z^T \dot{g}^* - Cg^T \dot{g}^* \end{aligned} \quad (77)$$

Here,

$$-k_p \|z\|^2 \quad \text{and} \quad -\frac{Ck_b}{d} \sin^2 \alpha$$

these are two negative terms, but we are not sure about the sign of

$$z^T P(g) g^*, \quad z^T \dot{g}^* \quad \text{and} \quad g^T \dot{g}^*.$$

These are all in the form

$$a^T b = \|a\| \|b\| \cos \theta$$

We do not know the angle θ between these components, so we cannot directly conclude if they are positive or negative.

So, for that we are going to use Cauchy–Schwarz inequality, Young’s inequality.

consider the first term $z^T P(g) g^*$, we have

$$|z^T P(g) g^*| \leq \|z\| \|P(g) g^*\| \quad (78)$$

$$\begin{aligned} \|P(g) g^*\|^2 &= (P(g) g^*)^T (P(g) g^*) \\ &= g^{*T} P(g)^T P(g) g^* \\ &= g^{*T} P(g)^2 g^* \\ &= g^{*T} P(g) g^* \\ &= g^{*T} (I - g g^T) g^* \\ &= 1 - (g^T g^*)^2 \\ &= 1 - \cos^2 \alpha \\ &= \sin^2 \alpha \end{aligned} \quad (79)$$

Therefore,

$$\|P(g)g^*\| = \sin \alpha \quad (80)$$

So,

$$|z^T P(g)g^*| \leq \|z\| \sin \alpha \quad (81)$$

Applying Young's inequality,

$$ab \leq \frac{\epsilon_1}{2}a^2 + \frac{1}{2\epsilon_1}b^2, \quad \epsilon_1 > 0 \quad (82)$$

Therefore,

$$|z^T P(g)g^*| \leq \frac{\epsilon_1}{2}\|z\|^2 + \frac{1}{2\epsilon_1}\sin^2 \alpha \quad (83)$$

Next term: $z^T \dot{g}^*$

$$|z^T \dot{g}^*| \leq \|z\| \|\dot{g}^*\| \quad (84)$$

Applying Young's inequality,

$$|z^T \dot{g}^*| \leq \frac{\epsilon_2}{2}\|z\|^2 + \frac{1}{2\epsilon_2}\|\dot{g}^*\|^2, \quad \epsilon_2 > 0 \quad (85)$$

Next term: $g^T \dot{g}^*$

$$|g^T \dot{g}^*| \leq \|g^T\| \|\dot{g}^*\| \quad (86)$$

Since

$$\|g\| = 1, \quad (87)$$

we have

$$|g^T \dot{g}^*| \leq \|\dot{g}^*\| \quad (88)$$

now, substituting these bounds into the derivative of the Lyapunov function, we get

$$\begin{aligned} \dot{V} \leq & -k_p\|z\|^2 + \frac{\epsilon_1}{2} \left(k_b + \frac{Ck_p}{d} \right) \|z\|^2 \\ & + \frac{1}{2\epsilon_1} \left(k_b + \frac{Ck_p}{d} \right) \sin^2 \alpha - \frac{Ck_b}{d} \sin^2 \alpha \\ & + \frac{d_{TL}\epsilon_2}{2} \|z\|^2 + \frac{d_{TL}}{2\epsilon_2} \|\dot{g}^*\|^2 + C\|\dot{g}^*\| \end{aligned} \quad (89)$$

$$\dot{V} \leq - \left[k_p - \frac{\epsilon_1}{2} \left(k_b + \frac{Ck_p}{d} \right) - \frac{d_{TL}\epsilon_2}{2} \right] \|z\|^2 \quad (90)$$

$$- \left[\frac{Ck_b}{d} - \frac{1}{2\epsilon_1} \left(k_b + \frac{Ck_p}{d} \right) \right] \sin^2 \alpha + \frac{d_{TL}}{2\epsilon_2} \|\dot{g}^*\|^2 + C\|\dot{g}^*\| \quad (91)$$

let,

$$\lambda_1 = k_p - \frac{\epsilon_1}{2} \left(k_b + \frac{Ck_p}{d} \right) - \frac{d_{TL}\epsilon_2}{2} \quad (92)$$

$$\lambda_2 = \frac{Ck_b}{d} - \frac{1}{2\epsilon_1} \left(k_b + \frac{Ck_p}{d} \right) \quad (93)$$

$$\dot{V} \leq -\lambda_1 \|z\|^2 - \lambda_2 \sin^2 \alpha + \frac{d_{TL}}{2\epsilon_2} \|\dot{g}^*\|^2 + C \|\dot{g}^*\| \quad (94)$$

6.0.1 Uniform Positivity of the Dissipation Coefficients

Recall that the coefficients in the Lyapunov derivative are given by

$$\lambda_1 = k_p - \frac{\epsilon_1}{2} \left(k_b + \frac{Ck_p}{d} \right) - \frac{d_{TL}\epsilon_2}{2}, \quad (95)$$

and

$$\lambda_2 = \frac{Ck_b}{d} - \frac{1}{2\epsilon_1} \left(k_b + \frac{Ck_p}{d} \right). \quad (96)$$

Under the separation assumption

$$d_{\min} \leq d(t) \leq d_{\max}, \quad 0 < d_{\min} < d_{\max} < \infty, \quad (97)$$

the factor $1/d(t)$ is uniformly bounded. In particular,

$$\frac{1}{d_{\max}} \leq \frac{1}{d(t)} \leq \frac{1}{d_{\min}}. \quad (98)$$

The controller gains and Young's inequality parameters are selected such that

$$\inf_{d \in [d_{\min}, d_{\max}]} \lambda_1(d) > 0, \quad \inf_{d \in [d_{\min}, d_{\max}]} \lambda_2(d) > 0. \quad (99)$$

Consequently, there exist constants

$$\underline{\lambda}_1 > 0, \quad \underline{\lambda}_2 > 0, \quad (100)$$

such that

$$\lambda_1(d) \geq \underline{\lambda}_1 > 0, \quad \lambda_2(d) \geq \underline{\lambda}_2 > 0 \quad (101)$$

for all admissible $d(t)$.

Hence, the dissipative terms in the Lyapunov derivative are uniformly negative over the admissible operating region.

Now define a function

$$\gamma(s) = \frac{d_{TL}}{2\epsilon_2}s^2 + Cs, \quad s \geq 0$$

$$d_{TL} > 0, \quad C > 0, \quad \epsilon_2 > 0$$

$$\gamma : \mathbb{R}_{\geq 0} \rightarrow \mathbb{R}_{\geq 0}$$

Properties

Continuity

Since

$$\gamma(s) = \frac{d_{TL}}{2\epsilon_2}s^2 + Cs$$

is a polynomial in the variable s , it is continuous for every

$$s \geq 0.$$

So, $\gamma(s)$ is continuous on $\mathbb{R}_{\geq 0}$. **Zero at the origin**

At $s = 0$,

$$\gamma(0) = 0 \tag{102}$$

Also,

$$\lim_{s \rightarrow \infty} \gamma(s) = \infty.$$

Strict monotonicity

$$\frac{d\gamma}{ds} = \frac{d_{TL}}{\epsilon_2}s + C \tag{103}$$

Since

$$d_{TL} > 0, \quad \epsilon_2 > 0, \quad C > 0$$

and

$$s \geq 0,$$

it follows that

$$\frac{d\gamma}{ds} = \frac{d_{TL}}{\epsilon_2}s + C > 0, \quad \forall s \geq 0. \tag{104}$$

So, $\gamma(s)$ is strictly increasing.

Positive definiteness

Since

$$\gamma(0) = 0 \tag{105}$$

and

$$\frac{d\gamma}{ds} > 0, \quad (106)$$

we have

$$\gamma(s) > 0, \quad \forall s > 0. \quad (107)$$

So, the function

$$\gamma(s) = \frac{d_{TL}}{2\epsilon_2} s^2 + C s \quad (108)$$

is continuous, satisfies $\gamma(0) = 0$, and is strictly increasing on $\mathbb{R}_{\geq 0}$. and also

$$\lim_{s \rightarrow \infty} \gamma(s) = \infty. \quad (109)$$

Hence,

$$\boxed{\gamma \in \mathcal{K}_\infty}$$

Therefore,

$$\gamma \in \mathcal{K}. \quad (110)$$

because

$$\mathcal{K}_\infty \subset \mathcal{K}. \quad (111)$$

So,

$$\dot{V} \leq -\lambda_1 \|z\|^2 - \lambda_2 \sin^2 \alpha + \gamma(\|\dot{g}^*\|) \quad (112)$$

where

$$\gamma(\|\dot{g}^*\|) = \frac{d_{TL}}{2\epsilon_2} \|\dot{g}^*\|^2 + C \|\dot{g}^*\|. \quad (113)$$

which is the standard ISS Lyapunov inequality required for the application of the input-to-state stability Lyapunov theorem.

And here, $\gamma(\|\dot{g}^*\|)$ is a class $\mathcal{K}$ function.

6.1 Input-to-State Stability (ISS)

Why ISS?

Classical Lyapunov theory considers autonomous systems of the form

$$\dot{x} = f(x) \quad (114)$$

If there exists a Lyapunov function $V(x)$ satisfying

$$V(x) > 0 \quad \text{for } x \neq 0 \quad (115)$$

$$V(x) = 0 \quad \text{for } x = 0 \quad (116)$$

and

$$\dot{V}(x) < 0, \quad (117)$$

then

$$x(t) \rightarrow 0 \quad (118)$$

which establishes asymptotic stability.

However, many practical control systems are continuously affected by external disturbances.

So, consider instead

$$\dot{x} = f(x, u) \quad (119)$$

where u represents the external disturbance.

ISS provides a mathematical analysis framework because it separates

- the intrinsic ability of the controller to reduce the tracking error,
- the unavoidable influence of the moving reference.

Since the desired reference evolves with time, the tracking error dynamics are continuously influenced by the motion of the reference trajectory.

Since the desired bearing $g^*(t)$ is time-varying, it is more appropriate to formulate the ISS analysis directly in terms of the tracking errors.

Define the bearing tracking error as

$$e_g = g - g^*. \quad (120)$$

The complete tracking-error vector is then defined as

$$e = \begin{bmatrix} z \\ e_g \end{bmatrix} = \begin{bmatrix} z \\ g - g^* \end{bmatrix}. \quad (121)$$

The time variation of the desired bearing is treated as the exogenous input

$$u = \dot{g}^*. \quad (122)$$

The Lyapunov inequality has the form

$$\dot{V} \leq -\lambda_1 \|z\|^2 - \lambda_2 \sin^2 \alpha + \gamma (\|\dot{g}^*\|). \quad (123)$$

Therefore, the controller continuously reduces both the position and bearing error, while the time variation of the desired bearing acts as an external excitation that limits how small the tracking error can become.

6.2 Input-to-State Stability Lyapunov Theorem

For the present tracking-error system, define the tracking-error state as

$$e = \begin{bmatrix} z \\ e_g \end{bmatrix}, \quad e_g = g - g^*, \quad (124)$$

and let

$$u = \dot{g}^*. \quad (125)$$

The error dynamics can therefore be written in the time-varying form

$$\dot{e} = F(e, u, t). \quad (126)$$

At zero tracking error and zero input, assume

$$F(0, 0, t) = 0, \quad \forall t \geq 0. \quad (127)$$

Assume that $F(e, u, t)$ is locally Lipschitz with respect to e on the admissible domain, uniformly with respect to time.

If there exists a continuously differentiable Lyapunov function $V(e)$ and functions

$$\alpha_1, \alpha_2, \alpha_3 \in \mathcal{K}_\infty, \quad \gamma \in \mathcal{K}, \quad (128)$$

such that

$$\alpha_1(\|e\|) \leq V(e) \leq \alpha_2(\|e\|), \quad (129)$$

and

$$\dot{V}(e) \leq -\alpha_3(\|e\|) + \gamma(\|u\|), \quad (130)$$

then the tracking-error system is input-to-state stable with respect to the input u on the region where these inequalities hold.

In the present analysis, because the bearing-error bound involving $\sin^2 \alpha$ is established only in a neighborhood of the desired bearing configuration, the resulting ISS property is local.

This Lyapunov formulation separates the stabilizing effect

$$-\alpha_3(\|e\|) \quad (131)$$

from the effect of the time-varying reference input

$$\gamma(\|u\|), \quad u = \dot{g}^*. \quad (132)$$

proof:-

Define the bearing tracking error as

$$e_g = g - g^*. \quad (133)$$

Therefore,

$$\dot{e}_g = \dot{g} - \dot{g}^*. \quad (134)$$

Since

$$\dot{g} = -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^*, \quad (135)$$

we obtain

$$\dot{e}_g = -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^* - \dot{g}^*. \quad (136)$$

Therefore, define the complete tracking-error vector as

$$e = \begin{bmatrix} z \\ e_g \end{bmatrix} = \begin{bmatrix} z \\ g - g^* \end{bmatrix}. \quad (137)$$

Using

$$\dot{z} = -k_p z + k_b P(g)g^* - d_{TL}\dot{g}^*, \quad (138)$$

the complete tracking-error dynamics are

$$\dot{e} = \begin{bmatrix} -k_p z + k_b P(g)g^* - d_{TL}u \\ -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^* - u \end{bmatrix}. \quad (139)$$

where

$$u = \dot{g}^*. \quad (140)$$

6.3 Verification of Local Lipschitz Continuity of the Error Dynamics

The closed-loop error dynamics are given by

$$\dot{z} = -k_p z + k_b P(g)g^* - d_{TL}\dot{g}^* \quad (141)$$

$$\dot{g} = -\frac{k_p}{d}P(g)z + \frac{k_b}{d}P(g)g^* \quad (142)$$

where

$$P(g) = I - gg^T. \quad (143)$$

To verify local Lipschitz continuity, each term appearing in the vector field is examined separately.

6.3.1 Linear Position Error Term

Consider the mapping

$$z \mapsto -k_p z. \quad (144)$$

Since multiplication by a constant is a linear operation, the mapping is linear.

Moreover, for any $z_1, z_2 \in \mathbb{R}^n$,

$$\| -k_p z_1 - (-k_p z_2) \| = k_p \| z_1 - z_2 \| \quad (145)$$

and therefore,

$$\| -k_p z_1 - (-k_p z_2) \| \leq L \| z_1 - z_2 \|, \quad (146)$$

where

$$L = k_p \quad (147)$$

is a Lipschitz constant.

So, $-k_p z$ is globally Lipschitz and therefore locally Lipschitz.

6.3.2 Projection Matrix

Consider the bearing vector

$$g = \begin{bmatrix} g_1 \\ g_2 \end{bmatrix}. \quad (148)$$

The projection matrix is

$$P(g) = I - gg^T \quad (149)$$

which, for $g \in \mathbb{R}^2$, becomes

$$P(g) = \begin{bmatrix} 1 - g_1^2 & -g_1 g_2 \\ -g_1 g_2 & 1 - g_2^2 \end{bmatrix}. \quad (150)$$

Each entry of the matrix consists of quadratic polynomials in the components of g .

Since polynomial functions are infinitely differentiable (C^∞), every entry of $P(g)$ is continuously differentiable.

A continuously differentiable mapping on an open set is locally Lipschitz. Therefore,

$$P(g) \quad (151)$$

is locally Lipschitz with respect to g .

6.4 Boundedness of the Bearing Control Term

Consider the nonlinear mapping

$$P(g)g^* \quad (152)$$

This expression is obtained by multiplying the continuously differentiable projection matrix $P(g)$ by the time-varying g^* .

Here, g^* is time varying, so we have to prove that it is bounded.

Proof

$$g^* = \frac{v_T}{\|v_T\|} \quad (153)$$

Differentiate with time,

$$\dot{g}^* = \frac{\dot{v}_T}{\|v_T\|} - \frac{v_T}{\|v_T\|^2} \frac{d}{dt} \|v_T\| \quad (154)$$

Since

$$a_T = \dot{v}_T, \quad (155)$$

we obtain

$$\dot{g}^* = \frac{a_T}{\|v_T\|} - \frac{v_T}{\|v_T\|^2} \frac{d}{dt} \|v_T\| \quad (156)$$

Also,

$$\|v_T\| = \sqrt{v_T^T v_T} \quad (157)$$

Therefore,

$$\frac{d}{dt} \|v_T\| = \frac{1}{2} (v_T^T v_T)^{-1/2} \frac{d}{dt} (v_T^T v_T) \quad (158)$$

$$= \frac{1}{2\|v_T\|} (\dot{v}_T^T v_T + v_T^T \dot{v}_T) \quad (159)$$

Since

$$\dot{v}_T = a_T \quad (160)$$

and

$$\dot{v}_T^T v_T = v_T^T \dot{v}_T, \quad (161)$$

as both are scalars,

$$\frac{d}{dt} \|v_T\| = \frac{v_T^T a_T}{\|v_T\|} \quad (162)$$

Thus,

$$\dot{g}^* = \frac{a_T}{\|v_T\|} - \frac{v_T v_T^T a_T}{\|v_T\|^3} \quad (163)$$

Since

$$g^* = \frac{v_T}{\|v_T\|}, \quad (164)$$

we have

$$\frac{v_T v_T^T}{\|v_T\|^2} = g^* g^{*T} \quad (165)$$

Therefore,

$$\dot{g}^* = \frac{1}{\|v_T\|} (I - g^* g^{*T}) a_T \quad (166)$$

$$= \frac{1}{\|v_T\|} P(g^*) a_T \quad (167)$$

Hence,

$$\|\dot{g}^*\| = \left\| \frac{1}{\|v_T\|} P(g^*) a_T \right\| \quad (168)$$

$$\leq \frac{1}{\|v_T\|} \|P(g^*)\| \|a_T\| \quad (169)$$

Since

$$\|P(g^*)\| = 1, \quad (170)$$

we obtain

$$\|\dot{g}^*\| \leq \frac{\|a_T\|}{\|v_T\|} \quad (171)$$

Now, under the assumption

$$\|a_T\| \leq a_{\max}, \quad \forall t \geq 0, \quad (172)$$

where $a_{\max} > 0$ is a finite constant, and

$$v_{\min} = \inf_{t \geq 0} \|v_T\| > 0, \quad (173)$$

we obtain

$$\|\dot{g}^*\| \leq \frac{a_{\max}}{v_{\min}} \quad (174)$$

This means if the target acceleration is uniformly bounded and the target speed remains bounded away from zero, then $\dot{g}^*$ is bounded.

Consequently, the desired bearing cannot change arbitrarily fast. This bound allows all terms involving $\dot{g}^*$ to be explicitly bounded and guarantees that the time-varying reference direction evolves smoothly.

Taken together, these properties show that the desired bearing vector is a well-defined, continuously differentiable, and bounded reference signal.

Consider the nonlinear mapping

$$g \longmapsto P(g)g^* \quad (175)$$

This expression is obtained by multiplying the continuously differentiable projection matrix $P(g)$ by the time-varying g^* .

Here, g^* is time varying, so we have to prove that it is bounded and that is already proved above.

Moreover, for any two bearing vectors g_1, g_2 ,

$$\begin{aligned}\|P(g_1)g^* - P(g_2)g^*\| &\leq \|P(g_1) - P(g_2)\|\|g^*\| \\ &\leq \|P(g_1) - P(g_2)\|,\end{aligned}\tag{176}$$

since

$$\|g^*\| = 1.\tag{177}$$

Now,

$$\begin{aligned}P(g_1) - P(g_2) &= (I - g_1g_1^T) - (I - g_2g_2^T) \\ &= g_2g_2^T - g_1g_1^T.\end{aligned}\tag{178}$$

Adding and subtracting $g_1g_2^T$,

$$\begin{aligned}P(g_1) - P(g_2) &= g_2g_2^T - g_1g_2^T + g_1g_2^T - g_1g_1^T \\ &= (g_2 - g_1)g_2^T + g_1(g_2 - g_1)^T.\end{aligned}\tag{179}$$

Therefore,

$$\begin{aligned}\|P(g_1) - P(g_2)\| &\leq \|(g_2 - g_1)g_2^T\| + \|g_1(g_2 - g_1)^T\| \\ &\leq \|g_2 - g_1\|\|g_2\| + \|g_1\|\|g_2 - g_1\|.\end{aligned}\tag{180}$$

Since g_1 and g_2 are unit bearing vectors,

$$\|g_1\| = \|g_2\| = 1,\tag{181}$$

hence

$$\|P(g_1) - P(g_2)\| \leq 2\|g_2 - g_1\|.\tag{182}$$

Therefore,

$$\|P(g_1)g^* - P(g_2)g^*\| \leq 2\|g_2 - g_1\|.\tag{183}$$

Hence, the mapping

$$g \longmapsto P(g)g^*\tag{184}$$

is globally Lipschitz with Lipschitz constant

$$L = 2.\tag{185}$$

The above derivation establishes an explicit Lipschitz bound for the nonlinear mapping

$g \mapsto P(g)g^*$. Since global Lipschitz continuity implies local Lipschitz continuity, the mapping is locally Lipschitz.

This result is stronger than merely stating that the mapping is locally Lipschitz.

6.4.1 Relative Bearing Dynamics

The second error equation contains the factor

$$\frac{1}{d} \tag{186}$$

where

$$d = \|p_T - p_L\|. \tag{187}$$

A singularity would occur if

$$d \rightarrow 0. \tag{188}$$

Assumption: The target and leader remain separated by a strictly positive distance, i.e.,

$$d = \|p_T - p_L\| \geq d_{\min} > 0,$$

over the interval of analysis.

Then,

$$\frac{1}{d} \leq \frac{1}{d_{\min}},$$

Since

$$d = \|p_T - p_L\| = \|z + d_{TL}g^*\|,$$

the distance is a Lipschitz function of the position tracking error. In particular,

$$|d_1 - d_2| \leq \|z_1 - z_2\|.$$

Under the separation assumption

$$d \geq d_{\min} > 0,$$

we have

$$\left| \frac{1}{d_1} - \frac{1}{d_2} \right| = \frac{|d_1 - d_2|}{d_1 d_2} \leq \frac{1}{d_{\min}^2} \|z_1 - z_2\|.$$

Therefore, $1/d$ is locally Lipschitz on the admissible region. Consequently, the mappings

$$\frac{1}{d}P(g)z \quad \text{and} \quad \frac{1}{d}P(g)g^*$$

are locally Lipschitz.

6.4.2 Input Term / Disturbance Term

The external input appearing in the leader tracking-error dynamics is

$$-d_{TL}\dot{g}^*. \quad (189)$$

Since the desired distance d_{TL} is constant, and the input

$$u = \dot{g}^* \quad (190)$$

is regarded as an exogenous input, the disturbance term can be written as

$$-d_{TL}u. \quad (191)$$

This term is independent of the tracking-error state

$$e = \begin{bmatrix} z \\ g - g^* \end{bmatrix}. \quad (192)$$

Therefore, for a fixed admissible input u , the mapping

$$e \mapsto -d_{TL}u \quad (193)$$

is constant with respect to the tracking-error state. Consequently, it does not affect the local Lipschitz continuity of the vector field with respect to e .

Combining the results obtained above, each component of the tracking-error vector field is either

- a linear mapping in the tracking error,
- a continuously differentiable nonlinear mapping,
- a product of locally Lipschitz mappings, or
- a state-independent input term.

Therefore, the complete tracking-error dynamics

$$\dot{e} = F(e, u, t) \quad (194)$$

are locally Lipschitz with respect to the tracking-error state e on the admissible domain, uniformly with respect to time, for every admissible bounded input u .

Since the external input

$$u = \dot{g}^* \quad (195)$$

is treated as an exogenous signal rather than as a state variable, local Lipschitz continuity with respect to u is not required for the present ISS analysis.

6.4.3 Comparison Bounds for the Lyapunov Function

The Lyapunov function is defined as

$$V = \frac{1}{2}\|z\|^2 + C(1 - g^T g^*), \quad C > 0. \quad (196)$$

Using the bearing tracking error

$$e_g = g - g^*, \quad (197)$$

the complete tracking-error vector is defined as

$$e = \begin{bmatrix} z \\ e_g \end{bmatrix} = \begin{bmatrix} z \\ g - g^* \end{bmatrix}. \quad (198)$$

and we have to proof that

$$\boxed{\alpha_1(\|e\|) \leq V \leq \alpha_2(\|e\|)}. \quad (199)$$

Proof

Since g and g^* are unit vectors,

$$1 - g^T g^* = \frac{1}{2}\|g - g^*\|^2. \quad (200)$$

Therefore, the Lyapunov function can be written as

$$V = \frac{1}{2}\|z\|^2 + \frac{C}{2}\|g - g^*\|^2. \quad (201)$$

Since

$$\|e\|^2 = \|z\|^2 + \|g - g^*\|^2, \quad (202)$$

define

$$m_1 = \min \left\{ \frac{1}{2}, \frac{C}{2} \right\}, \quad (203)$$

and

$$m_2 = \max \left\{ \frac{1}{2}, \frac{C}{2} \right\}. \quad (204)$$

Then,

$$V \geq m_1 (\|z\|^2 + \|g - g^*\|^2) \quad (205)$$

and therefore

$$V \geq m_1 \|e\|^2. \quad (206)$$

Similarly,

$$V \leq m_2 (\|z\|^2 + \|g - g^*\|^2), \quad (207)$$

which gives

$$V \leq m_2 \|e\|^2. \quad (208)$$

Now define the comparison functions

$$\alpha_1(r) = m_1 r^2, \quad r \geq 0, \quad (209)$$

and

$$\alpha_2(r) = m_2 r^2, \quad r \geq 0. \quad (210)$$

Since $m_1 > 0$ and $m_2 > 0$, both functions satisfy

$$\alpha_i(0) = 0, \quad (211)$$

are continuous and strictly increasing for $r \geq 0$, and satisfy

$$\lim_{r \rightarrow \infty} \alpha_i(r) = \infty, \quad i = 1, 2. \quad (212)$$

Hence,

$$\alpha_1, \alpha_2 \in \mathcal{K}_\infty. \quad (213)$$

Therefore, the required Lyapunov comparison bounds are

$$\boxed{\alpha_1(\|e\|) \leq V \leq \alpha_2(\|e\|)}. \quad (214)$$

6.4.4 Class $\mathcal{K}_\infty$ Comparison Functions

Now define the comparison functions

$$\alpha_1(r) = m_1 r^2, \quad (215)$$

and

$$\alpha_2(r) = m_2 r^2. \quad (216)$$

For all $r \geq 0$, since

$$m_1 > 0, \quad m_2 > 0, \quad (217)$$

both functions satisfy the following properties:

- $\alpha_i(0) = 0$,
- $\alpha_i(r)$ is continuous on $[0, \infty)$,
- $\alpha_i(r)$ is strictly increasing for $r \geq 0$,
-

$$\lim_{r \rightarrow \infty} \alpha_i(r) = \infty, \quad (218)$$

for $i = 1, 2$.

Hence,

$$\alpha_1, \alpha_2 \in \mathcal{K}_\infty. \quad (219)$$

Therefore, using the tracking-error vector

$$e = \begin{bmatrix} z \\ g - g^* \end{bmatrix}, \quad (220)$$

the Lyapunov function satisfies the comparison inequality

$$\boxed{\alpha_1(\|e\|) \leq V \leq \alpha_2(\|e\|)}. \quad (221)$$

Thus, the Lyapunov function is positive definite with respect to the tracking error e and satisfies the required comparison bounds on the admissible domain.

6.4.5 Proof of the ISS Lyapunov Derivative Inequality

We prove that the Lyapunov derivative satisfies

$$\dot{V}(e) \leq -\alpha_3(\|e\|) + \gamma(\|u\|). \quad (222)$$

where

$$u = \dot{g}^*. \quad (223)$$

From the previous result,

$$V = \frac{1}{2}\|z\|^2 + \frac{C}{2}\|g - g^*\|^2. \quad (224)$$

Also,

$$\|g - g^*\|^2 = 2(1 - g^T g^*) \quad (225)$$

and

$$g^T g^* = \cos \alpha. \quad (226)$$

Therefore,

$$1 - \cos \alpha = \frac{1}{2}\|g - g^*\|^2. \quad (227)$$

Now consider $\sin^2 \alpha$. Using

$$\sin^2 \alpha = 1 - \cos^2 \alpha \quad (228)$$

we obtain

$$\begin{aligned} \sin^2 \alpha &= (1 - \cos \alpha)(1 + \cos \alpha) \\ &= \frac{1}{2}\|g - g^*\|^2(1 + \cos \alpha). \end{aligned} \quad (229)$$

At the equilibrium,

$$g \rightarrow g^* \quad (230)$$

which means

$$\alpha \rightarrow 0, \quad \cos \alpha \rightarrow 1. \quad (231)$$

Hence, there exists a neighborhood of the equilibrium in which

$$1 \leq 1 + \cos \alpha \leq 2. \quad (232)$$

Multiplying by

$$\frac{1}{2} \|g - g^*\|^2,$$

we obtain

$$\frac{1}{2} \|g - g^*\|^2 \leq \frac{1}{2} \|g - g^*\|^2 (1 + \cos \alpha) \leq \|g - g^*\|^2. \quad (233)$$

Therefore,

$$\boxed{\frac{1}{2} \|g - g^*\|^2 \leq \sin^2 \alpha \leq \|g - g^*\|^2} \quad (234)$$

Now, from the previously obtained ISS inequality,

$$\dot{V} \leq -\lambda_1 \|z\|^2 - \lambda_2 \sin^2 \alpha + \gamma(\|u\|), \quad (235)$$

where $\gamma(\|u\|)$ is a class $\mathcal{K}$ function.

Using

$$\sin^2 \alpha \geq \frac{1}{2} \|g - g^*\|^2, \quad (236)$$

we obtain

$$\dot{V} \leq -\lambda_1 \|z\|^2 - \frac{\lambda_2}{2} \|g - g^*\|^2 + \gamma(\|u\|). \quad (237)$$

Define

$$\lambda = \min \left\{ \lambda_1, \frac{\lambda_2}{2} \right\} > 0. \quad (238)$$

Then,

$$\begin{aligned} \dot{V} &\leq -\lambda_1 \|z\|^2 - \frac{\lambda_2}{2} \|g - g^*\|^2 + \gamma(\|u\|) \\ &\leq -\lambda (\|z\|^2 + \|g - g^*\|^2) + \gamma(\|u\|). \end{aligned} \quad (239)$$

Since

$$e = \begin{bmatrix} z \\ g - g^* \end{bmatrix}, \quad (240)$$

we have

$$\|e\|^2 = \|z\|^2 + \|g - g^*\|^2. \quad (241)$$

therefore,

$$\dot{V} \leq -\lambda \|e\|^2 + \gamma(\|u\|). \quad (242)$$

Now define

$$\alpha_3(r) = \lambda r^2, \quad r \geq 0. \quad (243)$$

Since $\lambda > 0$, the function α_3 satisfies

$$\alpha_3(0) = 0, \quad (244)$$

$$\alpha_3(r) \text{ is continuous on } [0, \infty), \quad (245)$$

$$\alpha_3(r) \text{ is strictly increasing for } r \geq 0, \quad (246)$$

and

$$\lim_{r \rightarrow \infty} \alpha_3(r) = \infty. \quad (247)$$

Hence,

$$\boxed{\alpha_3 \in \mathcal{K}_\infty}. \quad (248)$$

Hence,

$$\alpha_3 \in \mathcal{K}_\infty. \quad (249)$$

Consequently, the Lyapunov derivative satisfies

$$\dot{V}(e) \leq -\alpha_3(\|e\|) + \gamma(\|u\|), \quad u = \dot{g}^*. \quad (250)$$

Local Nature of the Bearing-Error Bound

The lower bound used for the bearing-error term is

$$\sin^2 \alpha \geq \frac{1}{2} \|g - g^*\|^2. \quad (251)$$

This inequality follows from

$$\sin^2 \alpha = \frac{1}{2} \|g - g^*\|^2 (1 + \cos \alpha), \quad (252)$$

together with the condition

$$1 + \cos \alpha \geq 1. \quad (253)$$

The above condition is guaranteed in a neighborhood of the desired bearing configuration

$$\alpha = 0. \quad (254)$$

Hence, the lower bound on $\sin^2 \alpha$ is a *local* bound.

In particular, this inequality does not hold globally over the entire unit sphere. At

the antipodal configuration,

$$\alpha = \pi, \quad (255)$$

we have

$$\sin^2 \alpha = 0, \quad (256)$$

whereas

$$\|g - g^*\|^2 = 4. \quad (257)$$

Therefore,

$$0 \not\geq \frac{1}{2}(4) = 2. \quad (258)$$

Consequently, the dissipation inequality obtained using this lower bound is valid only locally around the desired bearing configuration. Therefore, the resulting stability conclusion must be stated as *local ISS* (or uniform local ISS when the time-varying reference and associated bounds are uniform), rather than global ISS. Since the above dissipation inequality is established in a neighborhood of the desired bearing configuration, the result is local. Moreover, the bounds are uniform with respect to time under the assumed bounds on $d(t)$, $a_T(t)$, and $v_T(t)$.

Therefore, the closed-loop tracking-error system is **uniformly locally input-to-state stable (ISS)** with respect to the exogenous input

$$u = \dot{g}^*. \quad (259)$$

Consequently, the Lyapunov derivative satisfies the standard ISS dissipation inequality

$$\dot{V} \leq -\alpha_3(\|e\|) + \gamma(\|u\|). \quad (260)$$

Therefore, the second hypothesis of the equilibrium ISS Lyapunov theorem is satisfied.

Therefore, under the stated separation, bounded-acceleration, and nonzero-target-speed assumptions, the closed-loop tracking-error system is uniformly locally input-to-state stable (ISS) with respect to the exogenous input

$$u = \dot{g}^*.$$

7 Conclusion

This M.Tech project presents the development and analysis of a **target–leader tracking control system** for a moving target. The system is formulated using a single-integrator model, and a desired reference position is generated for the leader based on a prescribed target–leader distance and desired bearing direction.

A control law incorporating **position feedback and bearing feedback** is developed to reduce the position tracking error and align the actual bearing vector with the desired bearing vector. The corresponding position and bearing error dynamics are derived to

obtain the complete closed-loop error system.

The stability of the proposed control system is investigated using a **composite Lyapunov function** that considers both position and bearing errors. The Lyapunov derivative is evaluated using Cauchy–Schwarz and Young’s inequalities, leading to conditions on the controller gains and system parameters under which the error dynamics exhibit dissipative behavior.

Furthermore, the effect of the **time-varying desired bearing** is considered by treating its rate of change as an external input. An **Input-to-State Stability (ISS)** framework is formulated to characterize the influence of this time-varying reference on the tracking errors. The analysis shows that the controller provides stabilizing behavior while the changing desired bearing acts as an external excitation.

Since the project is currently ongoing, the remaining work will focus on further **simulation, controller validation, performance evaluation, and refinement of the stability conditions**. The developed framework provides a mathematical foundation for achieving stable target–leader tracking while maintaining the desired distance and bearing relationship.